ONE-PAGE RESEARCH BRIEF | AUGUST 2026

Can a Physics-Guided Model Screen Ac-225 Production Plans Faster?

A student-built surrogate study for comparing irradiation and cooling conditions before trusted-solver verification.

Samuel Ogunnubi

12th-grade student researcher

Glen Burnie High School, Maryland

Biomedical and Allied Health

WHAT I AM TRYING TO DO

Ac-225 is promising for targeted alpha therapy, but production is difficult. I am testing whether a fast model can narrow many production plans, reject conditions it does not understand, and send only the strongest candidates to trusted nuclear calculations. It screens possibilities; it does not replace the physics.

HOW THE PROJECT WORKS

1 INPUTS

Beam energy, target geometry, irradiation time, and cooling time.

2 REACTION RATES

Predict the Ra-226(n,2n) and competing (n,gamma) rates.

3 ISOTOPE CHAIN

Track Ra-225 to Ac-225 and the Ra-227 to Ac-227 impurity branch.

4 SAFETY CHECK

Warn outside the training domain; verify accepted finalists with physics.

WHAT I BUILT AND TESTED

V3: physics-informed isotope-timeline model

- Reduced five-nuclide system with physics penalties and exact decay-only cooling checks.
- 1,400 Radau trajectories; 6,000 epochs; frozen seed-42 checkpoint and source hashes.
- Compared with a matched MLP, shifted tests, and a 10,000-schedule audit.

V4: geometry, uncertainty, and domain test

- GP versus five-seed LSTM on the same finite-geometry/TENDL proxy data.
- Split: 160 train, 24 select, 24 calibrate, and 48 locked test; paired-bootstrap ranking.

CURRENT RESULTS

1.83%
LSTM MEDIAN
p95: 3.93%

1.97%
GP MEDIAN
p95: 3.79%

95.8%
JOINT COVERAGE
calibrated 95%

100%
OOD REJECTION
184 of 184 warned

V3 endpoint test: 0.97% matched median; 1.57% shifted; matched MLP 2.91%.

Schedule audit: rank 2, 0.047% below Radau's best; 84.9x faster.

WHAT THESE RESULTS ACTUALLY MEAN

Both V4 models passed every predeclared gate, but the paired bootstrap did not resolve a winner. GP is only the provisional leader because its observed p95 was slightly lower. V4 is a static one-step geometry test, so it cannot prove that LSTM memory helps. The exact analytic baseline also remains faster for this small equation set.

WHAT FAILED

The Berkeley reality check

The first reconstruction of the published 33 and 40 MeV Ra-226 experiments fell outside V3's training support. Raw activity predictions were about 600 times too high.

I kept this failure instead of tuning the model to make two outside points look correct. The next test must distinguish a true model failure from missing experiment inputs or a reconstruction mismatch.

WHAT I NEED NOW

To run a frozen Berkeley external test without tuning, I need:

- Target-integrated neutron spectra or fluence for 33 and 40 MeV, including bins, units, and normalization.
- Beamspot and target-geometry correction inputs, plus beam-current or fluence history.
- Spectrum uncertainty, covariance, or parameter samples.
- A numerical Ac-227 detection limit, confidence level, and reference time, if available.

Claim boundary: Current evidence is modeled proxy data. It does not yet establish experimental, reactor, clinical, cost-savings, or universal under-3% performance.

Frozen artifacts, source hashes, full tables, failure analysis, and code are available.

sam.ogunnubi0@gmail.com